

Natural Computing

LMU Munich
summer term 2025

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*Another digression into
theoretical computer science???*

Who knows about
lazy evaluation?

Who has written an algorithm that
utilizes lazy evaluation?

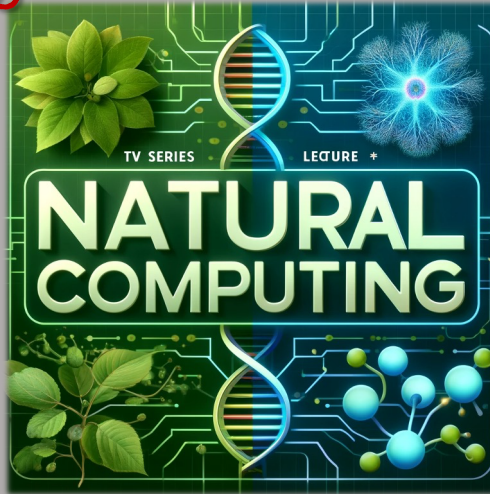
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lazy evaluation

eager/strict evaluation

And now for our main program...

Previously on...



two strange
quantum effects



Superposition



entanglement

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

ASCII letter A

$$q = \sum_{x \in \mathcal{X}} c_x |x\rangle$$

0	0	0	0	0	0	0	0
50%	50%	50%	50%	50%	50%	50%	50%
50%	50%	50%	50%	50%	50%	50%	50%
1	1	1	1	1	1	1	1

all ASCII chars
in superposition



measuring

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

ASCII char E with
probability 1/256

$$q = \sum_{x \in \mathcal{X}} c_x |x\rangle$$

0 90%	0 10%	0 90%	0 90%	0 90%	0 10%	0 90%	0 10%
10% 1	90% 1	10% 1	10% 1	10% 1	90% 1	10% 1	90% 1



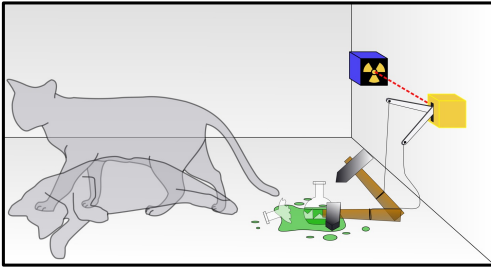
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

all ASCII chars
in superposition
(but probably E)

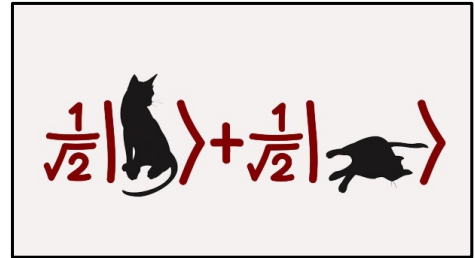
measuring

ASCII char E with
probability 0.43

Schrödinger's Cat



en.wikipedia.org/wiki/Schrödinger%27s_cat



www.motherjones.com/kevin-drum/2018/09/schrodingers-cat-is-alive-one-twelfth-of-the-time/

Schrödinger's Cat



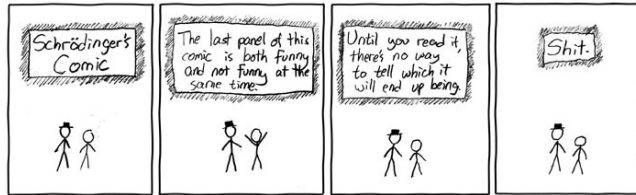
i.redd.it/9w2ifbjagtm41.jpg



i.redd.it/gq3ehlg8nfx71.jpg



tvtropes.org/pmwiki/pmwiki.php/UsefulNotes/SchrodingersCat

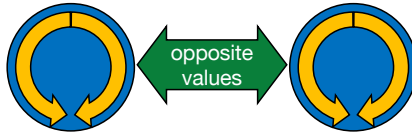


xkcd.com/45



i.redd.it/aky5jzorxu661.jpg

Entanglement



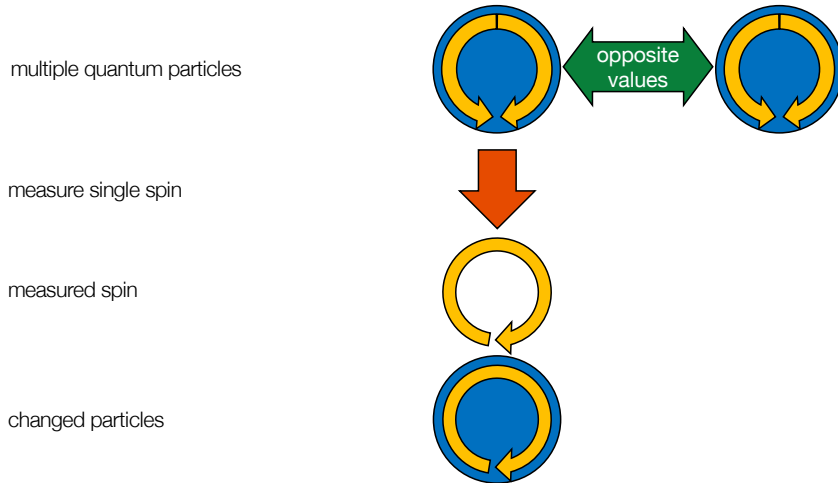
multiple quantum particles

measure single spin

measured spin

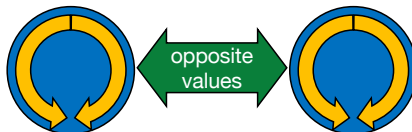
changed particles

Entanglement



Entanglement

multiple quantum particles



measure single spin



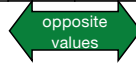
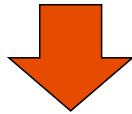
measured spin



changed particles



0	0	0	0	0	0	0	0
1	1	1	1	1	1	50% 1	50% 1



0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

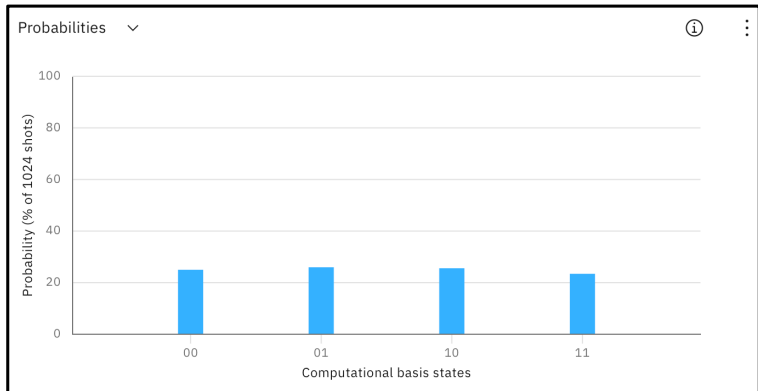
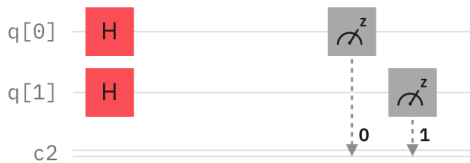
ASCII chars A B at
the same time

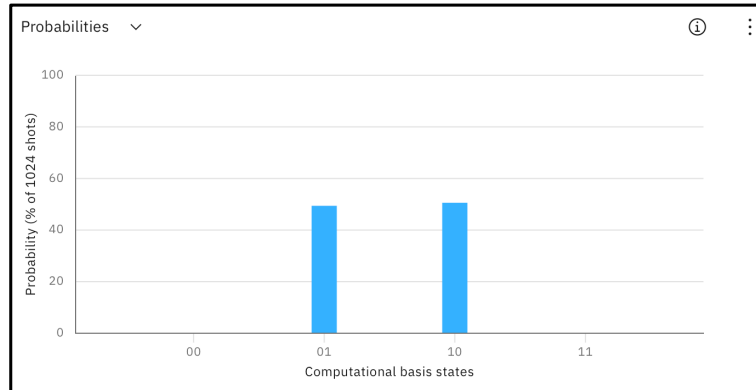
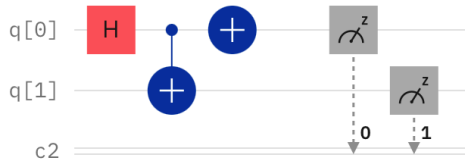
measuring

ASCII char A with
probability 1/2

Let's try

quantum-computing.ibm.com/composer





generated with quantum-computing.ibm.com/composer

Definition 6 (quantum gate). Let $q = \sum_{i=0}^{2^n-1} c_i |i\rangle$ be an n -qubit register. We often write

$$q = \begin{bmatrix} c_{0\dots 0} \\ \vdots \\ c_{1\dots 1} \end{bmatrix}$$

as a shorter notation when the basis $|0\dots 0\rangle, \dots, |1\dots 1\rangle$ is fixed.

An n -ary quantum gate is an operator that works on an n -qubit register q , i.e., (possibly) alters its superposition amplitudes, and thus produces a new n -qubit register q' . An n -ary quantum gate can be given via a matrix $U \in \mathbb{C}^{n \times n}$ so that

$$q' = Uq.$$

Quantum gates commonly implemented on quantum computers are unitary, which — among other things — implies they are reversible.

A little excursion:

What happens to computers
when they compute a lot?

A little excursion:

Why do computers get hot
when they compute?

A little excursion:

Why are most quantum operators
reversible?

Operator	Gate(s)	Matrix
Pauli-X (X)	 	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP	 	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$